

Trainings ▾

Preparatory Steps

With EGR

Remove the following:

- Fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).
- Cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/10/10-008-002.html>).
- Automatic belt tensioner (water pump). Refer to Procedure 008-080 (</qs3/pubsys2/xml/en/procedures/10/10-008-080.html>).

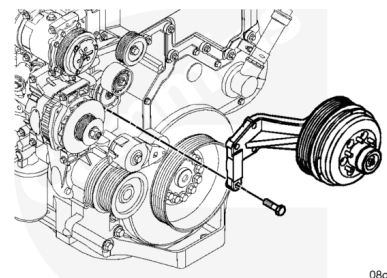
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Remove

With EGR

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



08c00146

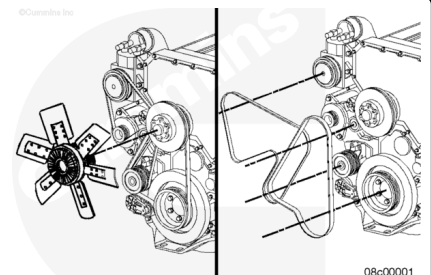
Remove the mounting capscrews and the fan hub assembly.

Automotive Applications Without EGR

Remove the fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).

Remove the alternator and water pump belts. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>) and Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).

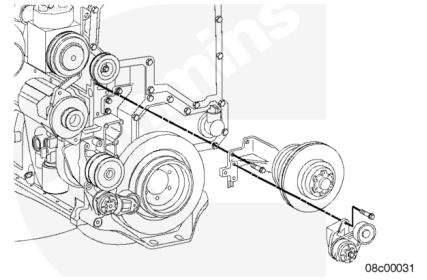
©Cummins Inc.



08c00001

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



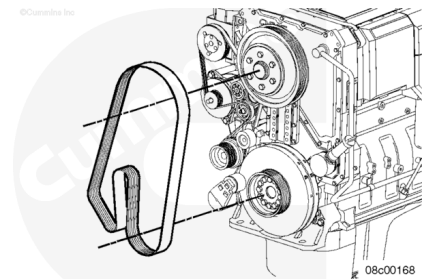
Remove alternator belt tensioner and bracket.

Remove the capscrews and the fan hub assembly.

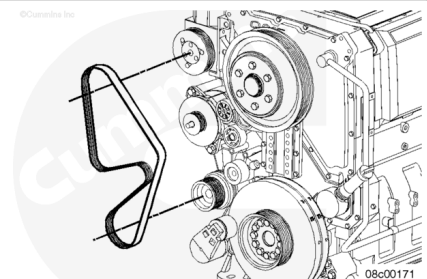
Industrial and Power Generation Application Without EGR

Remove the fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).

Remove the fan and water pump belt. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).

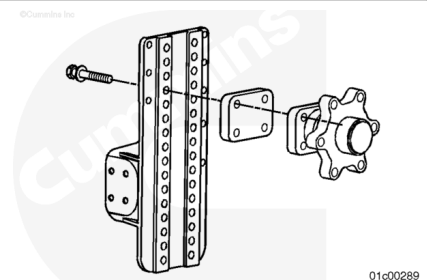


Remove the alternator drive belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).



Note : The fan support must be removed prior to removing the fan hub.

Remove the fan hub from the fan hub support. Refer to Procedure 008-088 (</qs3/pubsys2/xml/en/procedures/10/10-008-088.html>).



Clean and Inspect for Reuse

With EGR

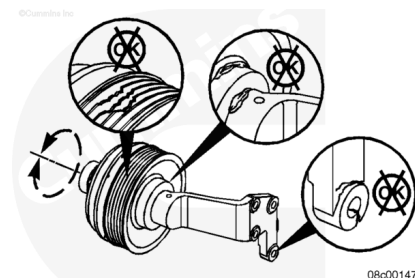
Inspect the fan hub for the following:

- Freedom of rotation

- Cracks
- Grease seal leakage.

Repair or replace the fan hub if the fan hub does **not** rotate freely or if there is evidence of cracks or grease seal leakage.

Refer to the OEM service literature for repair of the fan hub.

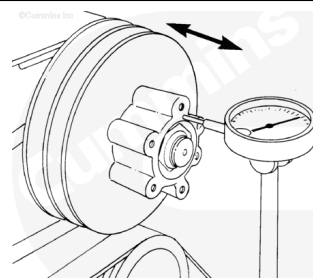


08c00147

Measure the fan hub end clearance play. The end clearance is dependent upon the fan hub style.

Replace the fan hub if the end clearance is **not** within the manufacturer's specifications.

Refer to OEM service manual for additional inspection steps.



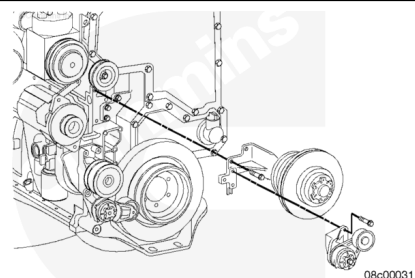
fa1bdta

Without EGR

Inspect the fan hub for the following:

- Freedom of rotation
- Cracks
- Grease seal leakage.

Repair or replace the fan hub if the fan hub does **not** rotate freely or if there is evidence of cracks or grease seal leakage.

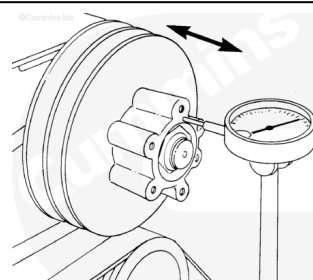


08c00031

Measure the fan hub end clearance. The end clearance is dependent upon the fan hub style.

Replace the fan hub if the end clearance is **not** within specifications.

mm		in
0.08	MIN	0.003
0.25	MAX	0.009



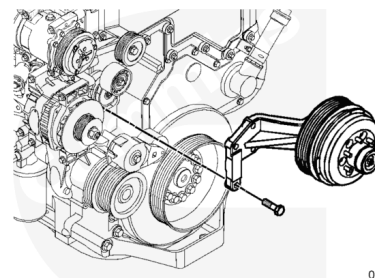
fa1bdta

Install

With EGR

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



Install the fan hub assembly and the capscrews.

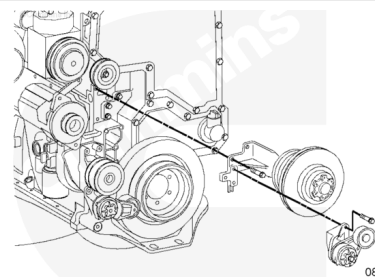
Tighten the capscrews.

Torque Value: 113 n•m [83 ft-lb]

Automotive Applications Without EGR

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



Install the fan hub and the capscrews.

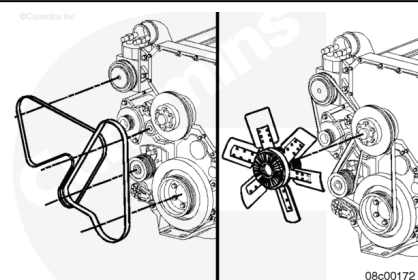
Tighten the capscrews.

Torque Value: 90 n•m [66 ft-lb]

Install the belt tensioner bracket.

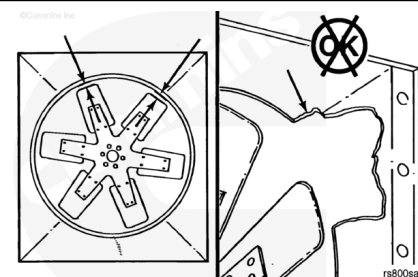
Install the belts. Refer to the Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>) and 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).

Install the fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).



Industrial and Power Generation Application Without EGR

Note : The fan hub must be installed in the correct location on the fan support to allow proper clearance between the fan and fan shroud. Refer to the OEM service manual.



Note : The fan hub must be installed prior to installing the fan support.

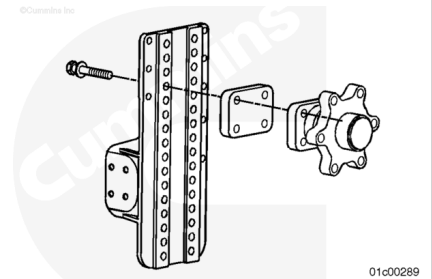
Install the fan hub on the fan support using four capscrews and washers.

Note : Make certain the correct fan center is used for the application.

Tighten the capscrews.

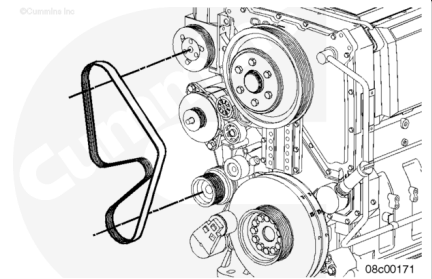
Torque Value: 90 n•m [66 ft-lb]

Install the fan support. Refer to Procedure 008-088 (</qs3/pubsys2/xml/en/procedures/10/10-008-088.html>).



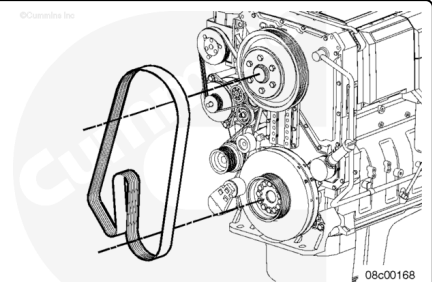
Install the alternator belt tensioner bracket.

Install the alternator drive belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).



Install the fan and water pump drive belt. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).

Install the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).



Finishing Steps

With EGR

Install the following:

- Automatic belt tensioner (water pump). Refer to Procedure 008-080 (</qs3/pubsys2/xml/en/procedures/10/10-008-080.html>).
- Cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/10/10-008-002.html>).
- Fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).

